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Variations in oral microbiome and its predictive functions between tumorous and healthy individuals

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Abstract

Introduction. The oral cavity is one of the largest reservoirs of microorganisms and many pathogenic bacteria have been shown to be associated with the aetiology of oral cancers.

Gap Statement. Owing to the complexity of oral microbial communities and their unclear relationship with oral cancer, identification of specific bacteria which contribute to oral cancer is a key imperative.

Aim. To compare and investigate the variations in the composition of the bacterial microbiome and its functions between patients with oral tumorous lesions and healthy subjects.

Methodology. Twenty-seven samples from individuals with oral tumours (five oral benign tumours and 22 oral squamous cell carcinomas) and 15 samples from healthy subjects were collected. Genomic DNA was extracted and the V3–V5 region of the 16S rRNA gene was sequenced. Subsequently, bioinformatic assessment was conducted using QIIME2, PICRUSt and linear discriminant analysis effect size analyses (LEfSe).

Results. The oral microbiota was composed mainly of the phyla *Proteobacteria* (31.76%, 35.00%), *Bacteroidetes* (30.13%, 25.13%) and *Firmicutes* (23.92%, 17.07%) in tumorous and healthy individuals, respectively. *Neisseria*, *Prevotella*, *Fusobacterium*, *Streptococcus*, *Capnocytophaga*, *Veillonella*, *Haemophilus*, *Prevotella*, *Porphyromonas* and *Leptotrichia* were the most abundant genera. Alpha diversity in the tumour group was significantly greater than that in the healthy group ($P<0.05$). Differential analysis of microbes between groups demonstrated a significantly higher number of *Neisseria*, *Veillonella*, *Streptococcus*, *Leptotrichia*, *Lautropia*, *Sphingopyxis*, *Sphingobium*, *Tannerella*, *Actinomyces* and *Rothia* in healthy controls compared with the tumour group. However, the genera *Treponema*, *Micrococcus*, *Pseudomonas*, *Janthinobacterium*, *Parvimos*, *Loktanella*, *Staphylococcus*, *Acinetobacter*, *Catonella*, *Aggregatibacter* and *Propionibacterium* were significantly higher in the tumour group. Pathways related to cancers, cell motility, environmental adaptation, metabolism and signal transduction were enhanced in the tumour group, while functions associated with immune system diseases, replication, repair and translation were significantly enhanced in the healthy group.

Conclusion. Variations in the oral microbiota and its functions showed a correlation with oral tumours. The tumour group showed an increased abundance of some multi-drug-resistant and periodontitis-related pathogens. The significantly altered microbiotas may serve as potential biomarkers or inform combination therapy for oral tumours.

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Keywords: oral tumours; microbiome; microbial functions; NGS.

Abbreviations: ASV, amplicon sequence variant; FDR, false discovery rate; FEP, fibroepithelial polyp; LDA, linear discriminate analysis; LEfSe, linear discriminant analysis effect size analysis; OSCC, oral squamous cell carcinoma; OSMF, oral submucosa fibrosis; PCoA, principal coordinates analysis; TCS, two-component sensing system.

All sequencing results were submitted to NCBI with PRJNA730970 (SRR14583850–SRR14583891).

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Three supplementary table is available with the online version of this article.

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INTRODUCTION

Head and neck cancers are among the most common cancers worldwide [1], accounting for approximately 5% of all tumours [2]. Approximately half of all head and neck cancers occur in the oral cavity [2]. An estimated 354864 newly diagnosed cases of lip and oral cavity cancers were reported in 2018, and these sites ranked 18th among the most commonly encountered cancer sites [1]. In the USA, oral cancer was regarded as the sixth most common cause of cancer-related mortality [2] and 52260 new cases of oropharyngeal cancers were diagnosed in 2020 [3]. Oral squamous cell carcinomas (OSCCs) are considered to be the most common malignant oral tumours [1]. The financial cost for managing oral cancer is considered to be the highest among all cancers [4]. In addition, oral carcinoma has a high prevalence in Southern Asia, because of widespread use of tobacco-related products in this region [5]. In 2015, an estimated 51765 newly diagnosed cases of oral or oropharyngeal cancers were reported in China, and 23830 deaths were attributable to oral cancer [6].

In-depth characterization of the causes of oral cancer can help inform preventive interventions. Chronic use of alcohol, tobacco and betel quid chewing are the most important risk factors for oral cancer [7–9]. However, in a recent study by Bai *et al*., smoking and alcohol consumption were not found to be prognostic factors for oral cancer [10], which suggested the possibility of additional aetiological factors for oral tumours. In the 1990s, *Helicobacter pylori* was identified as the main causative agent for the development of gastric cancer [11]. Subsequent research revealed an association of human cancers with certain viruses, bacteria and parasites, and several microbes were identified as strong risk factors for specific types of cancers [12–14]. The oral cavity is considered as one of the largest reservoirs of different types of microorganisms, including different species of bacteria, fungi and viruses [15]. Studies have suggested an aetiological role of many pathogenic bacteria in the causation of oral cancers [16]. *Porphyromonas gingivalis* and *Fusobacteria* species were more prevalent on the surface of OSCCs [17]. Al-hebshi *et al.* suggested a close association of *Fusobacterium nucleatum* and *Pseudomonas aeruginosa* with OSSC [18]. In addition, Chen *et al.* unravelled the variations in the composition of the salivary microbiome between patients with OSCC and oral submucosa fibrosis (OSMF) and further found considerable reduction in species and phylogenetic diversity in OSMF patients compared with those with OSCC [19]. Recently, Yang *et al.* found a close association of the oral microbiota community dynamics with OSCC stage; in addition, microbiota communities in OSCC patients from stage 4 showed significantly greater complexity compared to that from healthy controls [16]. Moreover, they found a significant increase in the abundance of *Fusobacteria* with the progression of oral cancer; however, the abundance of *Streptococcus*, *Haemophilus*, *Porphyromonas* and *Actinomyces* decreased with the progression of oral cancer [16]. In another study, the abundances of *Bacillus*, *Enterococcus*, *Parvimonas*, *Peptostreptococcus* and *Slackia* were different between patients with premalignant lesions and OSCC [20]. Schmidt *et al.* found significant overrepresentation of *Streptococcus* and *Rothia* at cancer lesion sites compared with the contralateral part of the anatomical site containing normal tissues in patients with oral cancer [21]; however, other studies found no association between the presence of individual subgingival pathogens and risk of cancer [22]. Owing to the complexity of oral microbial communities and the conflicting results of their relationship with oral cancer, identification of specific bacteria that contribute to oral cancer is a key imperative.

A wide range of technologies have been adopted by researchers to assess the correlation of various microbiota with oral cancer, such as pan-pathogen array [23], quantitative PCR [24], immunological staining [25], antibody detection [26] and fluorescence *in situ* hybridization [27]. However, these methods either depend on bacterial culture or have low throughput. Of note, the vast majority of the microbiota cannot be cultured [28]; therefore, these methods cannot elaborate on the profiles of the oral microbiota. High-throughput sequencing technology, which is independent of culture, can identify thousands of microorganisms by partial sequencing of the 16S rRNA gene.

To date, there is limited literature on the microbial variations between oral cancerous and healthy subjects. Therefore, in this study, we aimed to compare the characteristics of oral microbiotas between patients with oral tumours and healthy subjects. In addition, we compared the variations in the microbiome and its functions in patients with oral tumours (including benign tumours and OSCCs) and in healthy subjects by high-throughput sequencing technology.

METHODS

Tumour diagnosis was confirmed by histopathology. Patients who were diagnosed with oral cancer for the first time and had no previous history of any type of tumour were included in the study. Patients who received antibiotics for ≥2 weeks before this study were excluded from the study group. Healthy control samples were carefully examined clinically and specialized examination was carried out to exclude the presence of any lesion; healthy controls had no history of other oral mucosal diseases or severe systemic disorders. Based on a well-defined clinical protocol, patients with a history of betel chewing, diabetes mellitus or other systemic complications, such as heart or liver failure, were excluded from this study.

Written informed consent was obtained from all participants. All experiments were performed according to the approved guidelines and regulations of this institution.

The surface of benign tumour tissues, OSCC tissues and identical tissues from healthy subjects were sampled using two identical sterile swabs. Disposable sterile nylon flocking swabs were used (cy-98000; Hua Chen Yang) and then stored in oral swab

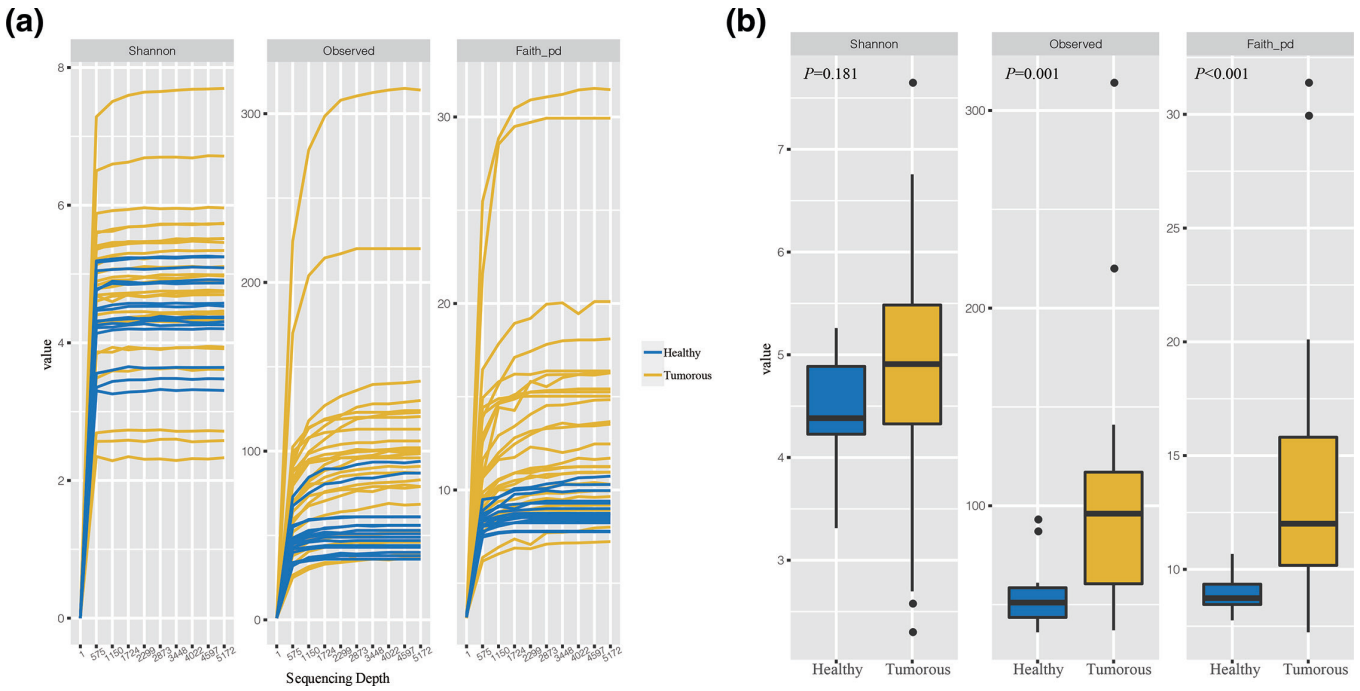


Fig. 1. Rarefaction curves and boxplots of α-diversity in the healthy and tumorous groups. The healthy controls had no history of other oral mucosal diseases or severe systemic disorders; the tumours from the patients were confirmed by histopathology. Observed species, Shannon indices and Faith's PD diversity were calculated using QIIME2, and then plotted using ggplot2. The plateau rarefaction curves indicated that the quenching depth covered the maximum of the diversity per sample. The index of phylogenetic diversity (P<0.001, Mann–Whitney U test) and observed species (P=0.001, Mann–Whitney U test) indicated that the diversity of the bacterial communities in the tumorous group was significantly greater than that in the healthy control group. The Shannon index showed higher bacterial communities in tumorous subjects compared with healthy individuals; however, the difference was not statistically significant (P>0.1, Mann–Whitney U test).

preservation solution (Tris, EDTA and antiseptic) to prevent DNA degradation. Sterile swabs without sampling were used as controls. All samples were stored at 4 °C and transferred to an ultra-low-temperature freezer (–80 °C) until processing.

DNA extraction and sequencing

Genomic DNA from the oral samples were extracted using the power soil DNA isolation kit (MoBio Laboratories) [29]. Each identical swab was cut and placed into individual tubes. Then, lysis buffer was added to the tubes, which were heated to 65 °C for 10 min and shaken for 20 min in a MoBio 96-well plate shaker (MoBio Laboratories). All steps were performed according to the manufacturer's instructions. Two parallel DNA samples from one patient were pooled into one tube to minimize any potential DNA extraction bias. After amplifying the variable V3–V5 region of the 16S rRNA gene with primers 515F/907R, the amplicon libraries were constructed. Finally, Illumina high-throughput paired-end sequencing was performed using an Illumina PE250 platform (Illumina).

Bioinformatics and statistical analysis

Bioinformatics analysis and assessment were conducted using the DADA2 pipeline in QIIME2 [30]. The reads were clustered into amplicon sequence variants (ASVs) and taxonomy assignment was performed as previously published [31, 32]. Analysis of alpha (α) microbial diversity was performed using QIIME2 and further plotted using the ggplot2 package in R software. A Mann–Whitney U test or *t*-test was conducted to detect significant differences in α-diversity. Principal coordinates analysis (PCoA) was performed to assess the differences in microbial distribution between the groups. Permutational multivariate analysis of variance (PERMANOVA) was applied to further compare the variability in genus abundance and compositions of microbiota between the groups. The Vegan package 2.0–10 in R software (version 3.2) was used to conduct PCoA. Linear discriminant analysis effect size analysis (LEfSe) was performed to compare the relative number of different taxa between the groups [33]. The functions of the microbial communities were predicted using PICRUSTt (v1.1.1) [34]. Then, STAMP (v2.1.3) was applied to determine the functional variations between tumorous and healthy subjects. P values of <0.05 were considered indicative of statistical significance.

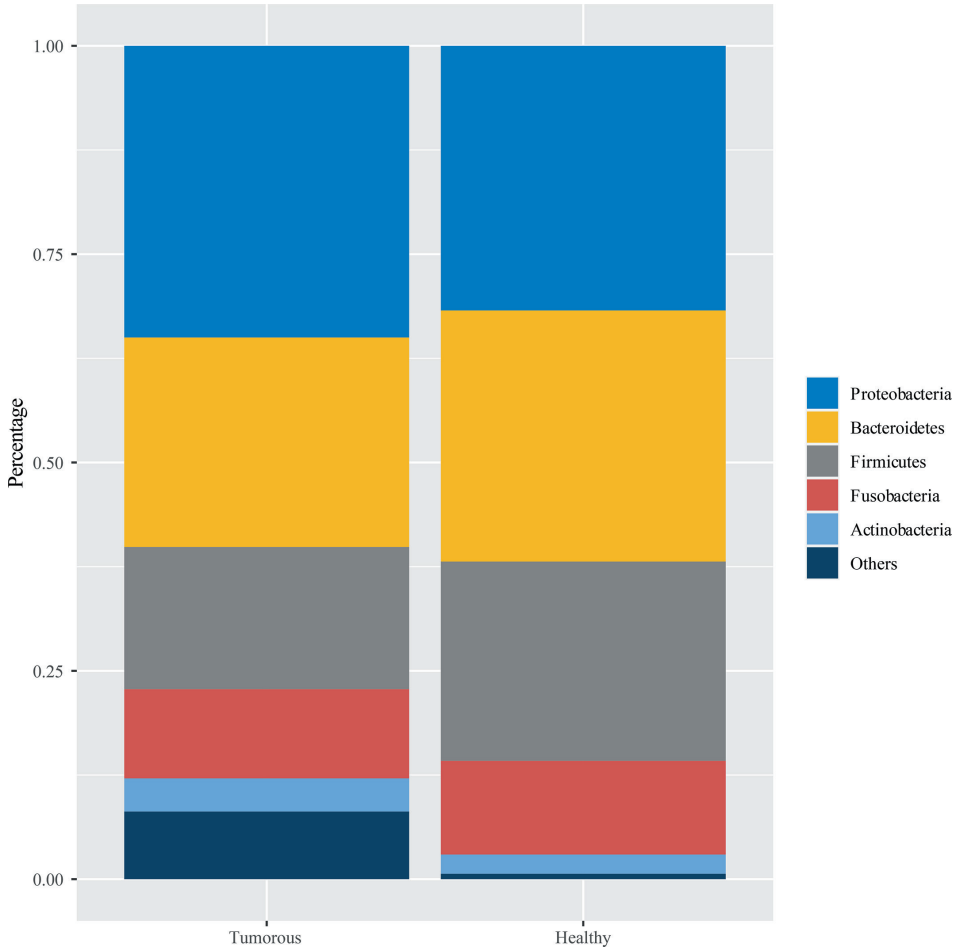


Fig. 2. The most abundant phyla between tumorous and healthy control samples. The healthy controls had no history of other oral mucosal diseases or severe systemic disorders; the tumours from the patients were confirmed by histopathology. A total of 32 phyla were identified in the two groups. Among these, 10 were discovered in healthy subjects and 32 were discovered in patients with tumours. In this plot, only the five most abundant phyla are shown, and the rest are classified as others.

RESULTS

Sampling and sequencing

The details of the 42 swabs collected, which included 27 swabs from tumorous patients and 15 from healthy controls, are presented in Table S1 (available in the online version of this article). There were no significant between-group differences with respect to age (51.41±15.37 versus 46.07±11.09 years, t=1.2975, P=0.2025, t-test) or sex [Chi-squared test (χ^2)=0.0021605, P=0.9629]. After genomic DNA extraction and sequencing, 1853614 paired reads were obtained. Following quality control and filtration with the DADA2 pipeline, 519 545 reads were produced with an average of 12370 reads per sample. A total of 1901 features were identified in the samples (Table S2). No nucleotides were detected in controls. All sequencing results were submitted to NCBI with numbers SRR14583850–SRR14583891.

Diversity and variations in microbial community between different groups

Different indices (e.g. richness, Shannon and Faith's PD) were used to assess the α-diversity of the bacterial community (Fig. 1). Plateau rarefaction curves indicated that the quenching depth covered the maximum of the diversity per sample and new ASVs with additional sequencing would not be expected. The α-diversity indicated a significant difference between the two groups. The phylogenetic diversity index indicated that the diversity of the bacterial communities in the tumorous group was significantly greater than that in the healthy control group (P<0.001, Mann–Whitney U test). A similar trend was demonstrated with respect to the observed species (P=0.001, Mann–Whitney U test). In addition, the Shannon index showed that there were higher bacterial communities in tumorous subjects compared with healthy individuals; however, the difference was not statistically significant (P>0.1, Mann–Whitney U test).

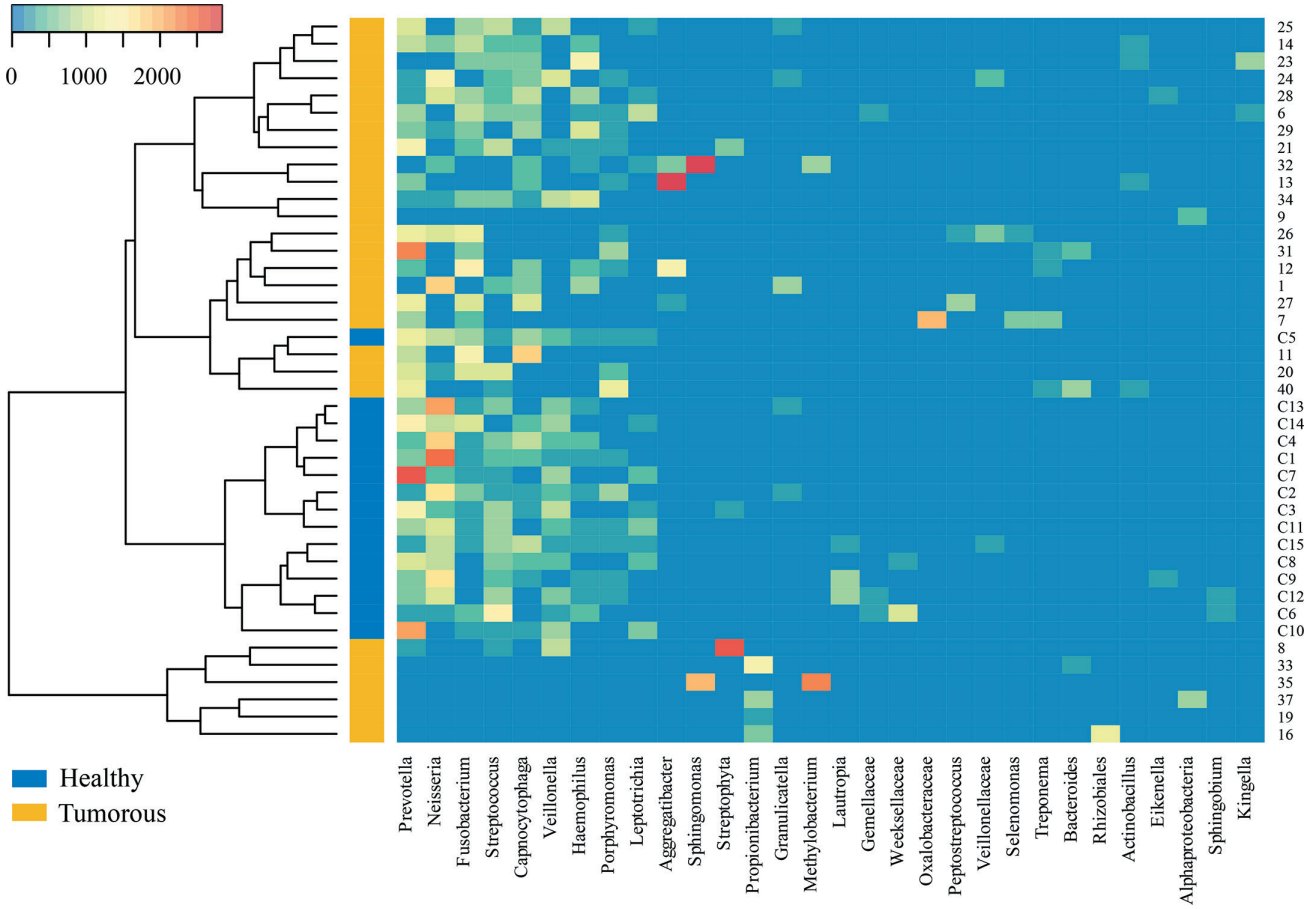


Fig. 3. Heatmap and hierarchical clustering of the top 30 genera in all samples. The dendrogram was created based on the Euclidean distance and Ward distance. D2 clustering methods were performed in heatmap packages in R software. The genera *Neisseria*, *Prevotella*, *Fusobacterium*, *Streptococcus*, *Capnocytophaga*, *Veillonella*, *Haemophilus*, *Porphyromonas*, *Leptotrichia* and *Aggregatibacter* were the most abundant in all the groups. Hierarchical clustering suggested that the samples from the tumorous group were clustered differently from those collected from the healthy controls.

Taxonomic analysis at the phylum level

A total of 32 phyla were identified in the two groups (Table S3). Among these, 10 were discovered in healthy subjects and 32 were discovered in tumorous patients. In particular, 22 phyla were only discovered in the tumorous group, including *Acidobacteria*, *Chloroflexi*, *Chlorobi*, *Planctomycetes*, *Thermi*, *Nitrospirae*, *Synergistetes*, *OP8*, *Verrucomicrobia*, *Gemmatimonadetes*, *OP3*, *WS1*, *Microgenomates*, *Lentisphaerae*, *AC1*, *Elusimicrobia*, *WS3*, *TPD.58*, *Parcubacteria*, *TM7*, *NC10* and *Fibrobacteres*. Among the 32 phyla, the five most abundant phyla in the tumorous and healthy controls accounted for 99.35 and 91.89% of the total microbes, respectively (Fig. 2). In the healthy controls, *Proteobacteria* was the most predominant phylum, which accounted for 31.76%. The second most abundant phylum was *Bacteroidetes*. which accounted for 30.13%. The next most commonly detected phyla were *Firmicutes* at 23.92%, *Fusobacteria* at 11.24%, *Actinobacteria* at 2.31% and others at 0.65%. A similar order of phylum abundance was revealed in the tumorous group. However, the abundance of *Proteobacteria* was higher in the tumorous group compared with the healthy controls. The percentage of *Proteobacteria* was higher (35.00%), while that of *Bacteroidetes* was lower (25.13%) in the tumorous group. The phyla *Firmicutes* and *Fusobacteria* were slightly reduced (17.07 and 10.70, respectively), while the percentage of *Actinobacteria* was higher (3.98%) in the tumorous group. The top five phyla accounted for 91.88% and the others accounted for 8.12%, which was much higher than that in the healthy control group (0.65%). This suggested that the microenvironment in tumorous patients had denser cultures with different types of microbiota.

Taxonomic analysis at the genus level

At the genus level, a total of 207 different genera were identified. The top 30 genera among all the samples were selected and are shown in Fig. 3. The genera *Neisseria*, *Prevotella*, *Fusobacterium*, *Streptococcus*, *Capnocytophaga*, *Veillonella*, *Haemophilus*, *Porphyromonas*, *Leptotrichia* and *Aggregatibacter* were the most abundant in all the groups. Hierarchical clustering suggested that the samples from the tumorous group were clustered differently from those collected from the healthy controls.

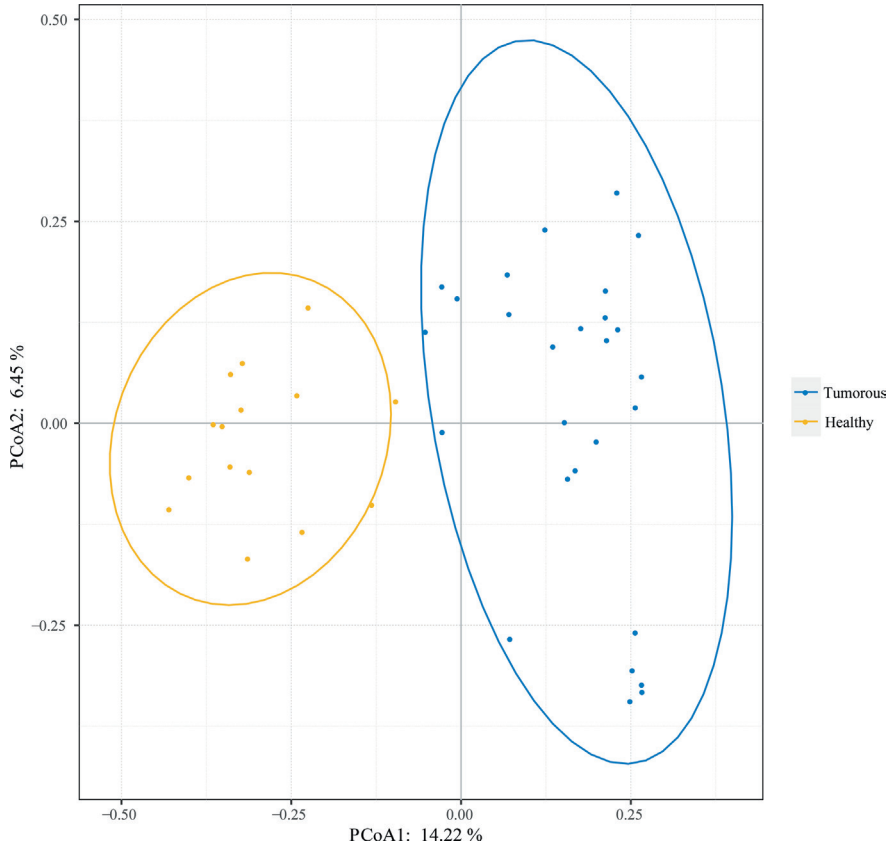


Fig. 4. Principal component analysis (PCoA) at the ASV level. PCoA was conducted using QIIME2, then plotted in R software using ggplot2. The ellipse was plotted with a confidence level of 0.95. The spots that represent the tumorous samples show a more dispersed distribution pattern than those from the healthy controls. Samples from the tumorous group were significantly different from those in the control group (PERMANOVA, pseudo-*F*=5.756921, *P*=0.001).

Neisseria was the most dominant genus in the healthy control group, and accounted for 21.23%, followed by *Prevotella* (17.57%), *Streptococcus* (9.74%), *Veillonella* (9.09%) and *Fusobacterium* (6.70%). However, in the tumorous group, the most abundant genera were *Prevotella* (11.02%), *Fusobacterium* (9.02%), *Capnocytophaga* (6.62%), *Neisseria* (5.42%) and *Haemophilus* (4.93%) (Fig. 3). Of interest, the top 10 genera in the control group accounted for 84.77%, while the corresponding percentage in the tumorous group was significantly lower (55.08%). The top 30 genera decreased dramatically from 98.43 to 78.56%. This indicated that the microbiota in the tumorous group was more evenly distributed compared with that in the healthy control group.

Microbial signatures associated with tumorous and control group

Beta diversity revealed the differences in the composition of the bacterial community between the two groups. PCoA at the genus level was used to determine the distribution of microbiota between the two groups (Fig. 4). As indicated by the hierarchical clustering (Fig. 3), the spots that represent the tumorous samples showed a more dispersed distribution pattern than those from the healthy controls. Samples from the tumorous group were significantly different from those in the control group (PERMANOVA, pseudo-*F*=5.756921, *P*=0.001).

The LEfSe method was used to identify the differential expression of genera between the two groups (Fig. 5). The genera *Neisseria*, *Veillonella*, *Streptococcus*, *Leptotrichia*, *Lautropia*, *Sphingopyxis*, *Sphingobium*, *Tranerella*, *Actinomyces* and *Rothia* were significantly higher in the healthy control group. However, *Treponema*, *Micrococcus*, *Pseudomonas*, *Janthinobacterium*, *Parvimos*, *Loktanella*, *Staphylococcus*, *Acinetobacter*, *Catonella*, *Aggregatibacter* and *Propionibacterium* were significantly higher in the tumorous group.

Differential oral microbial community function

PICRUSt and STAMP revealed significant functional divergences in several genes involved in different pathways (Figs 6 and 7). Similar to the distribution of microbial taxa, the distribution of gene functions was significantly different between the tumorous and healthy control groups [*t*-test, false discovery rate (FDR) correction, corrected *P*<0.05]. The functions connected with cancers

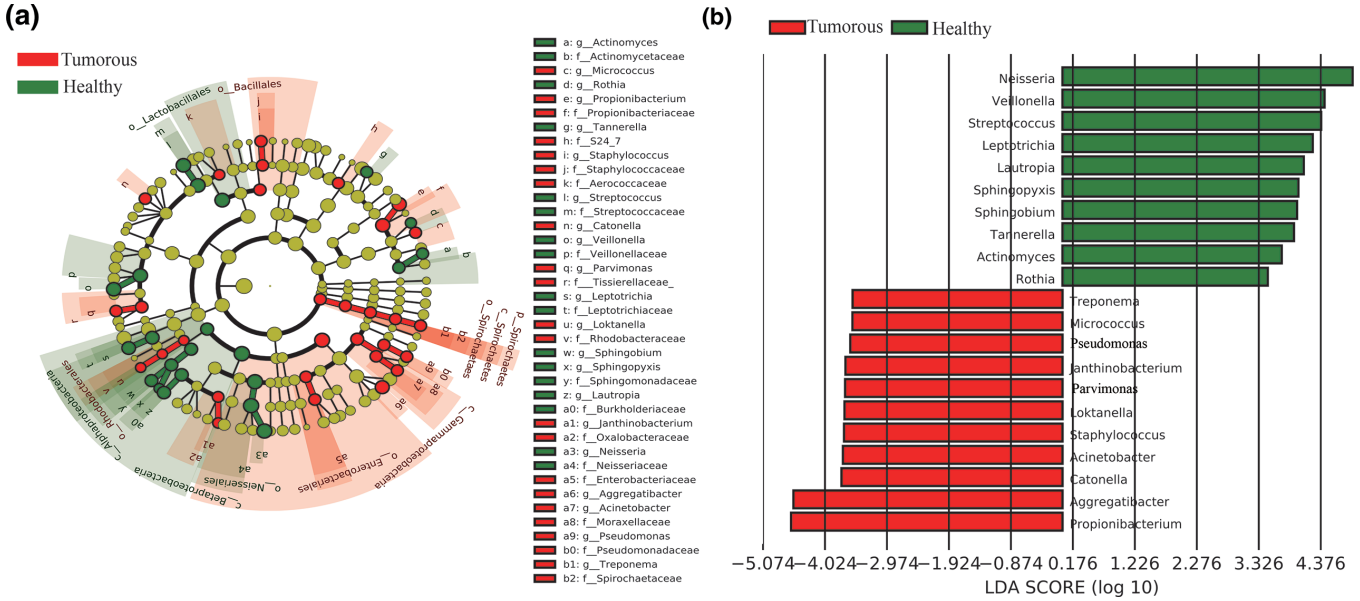


Fig. 5. Distinct taxa identified between two groups using linear discriminant analysis effect size analyses (LEfSe). The results of linear discriminate analysis (LDA) showed that the bacteria were significantly different between the tumorous and healthy control samples: (a) Cladogram indicating the phylogenetic distribution of bacteria that were significantly increased; and (b) LDA scores showing significant bacterial differences between groups at the genus level. The genera *Neisseria*, *Veillonella*, *Streptococcus*, *Leptotrichia*, *Lautropia*, *Sphingopyxis*, *Sphingobium*, *Tranerella*, *Actinomyces* and *Rothia* were significantly higher in the healthy control group, whereas *Treponema*, *Micrococcus*, *Pseudomonas*, *Janthinobacterium*, *Parvimos*, *Loktanella*, *Staphylococcus*, *Acinetobacter*, *Catonella*, *Aggregatibacter* and *Propionibacterium* were significantly higher in the tumorous group.

(corrected P=0.010), cell motility (corrected P=0.009), environmental adaptation (corrected P<0.001), metabolism (corrected P<0.001) and signal transduction (corrected P<0.011) were significantly enhanced in the tumorous group compared with the control healthy group. However, the pathways related to immune system diseases (corrected P=0.013), replication and repair (corrected P=0.021), and translation (corrected P=0.028) were significantly enhanced in the control healthy group (Fig. 6).

On level 3, 27 different pathways were significantly different (Fig. 7). Among these, 17 were more abundant in the tumorous group and 10 were were more abundant in the control healthy group (*t*-test, Storey FDR correction, corrected P<0.05). The pathways that were related to MAPK signalling, two-component system, bacterial chemotaxis, flagellar assembly, sulphur relay systems and cell motility were significantly increased in the tumorous group compared with the healthy control group. However, the samples from the control group had more genes related to lysine biosynthesis and aromatic amino acids (phenylalanine, tyrosine and tryptophan) biosynthesis.

DISCUSSION

There are approximately 1000 species that are capable of achieving stable colonization in the oral cavity [35]. The oral microflora is regarded as the second most complex microbiota in humans [36]. The development of high-throughput DNA sequencing technology at the molecular genetics level has facilitated in-depth characterization of the most important aspects of the oral microbiota. In addition, it has helped identify the oral microbiota associated with oral health and oral cancer [37]. However, the results pertaining to the relationship between oral health and specific microbiota have been inconsistent, because of diverse human lineages, such as related to different living conditions, diverse cultural backgrounds and different dietary habits [38]. Use of different technology [39], diverse types of sample [40, 41] and various methods of sequencing [42] may have contributed to the inconsistent results. In this study, the surfaces of tumorous lesions and healthy controls were sampled and assessed further. Oral samples from tumorous and healthy individuals were collected. The V3–V5 region of the 16S rRNA gene was chosen for sequencing and analysis [43].

Our results indicated greater α-diversity in species in the tumorous group compared with the healthy control group (Fig. 1), which was consistent with previous studies [40, 44]. In a detailed large-scale assessment of the salivary microbiota conducted by Yoshihisa *et al.*, the bacterial richness in the salivary microbiota was found to be significantly associated with poor oral health [38]. In general, good oral health was associated with lower diversity in the salivary microbiome [45]. A recent survey of *in situ* bacterial effects indicated that patients with a high microbiome diversity had increased tumour infiltration of CD4+ lymphocytes, which consequently prolongs the management of cancer [46]. Heshiki *et al.* examined the influence of the microbiome on the treatment

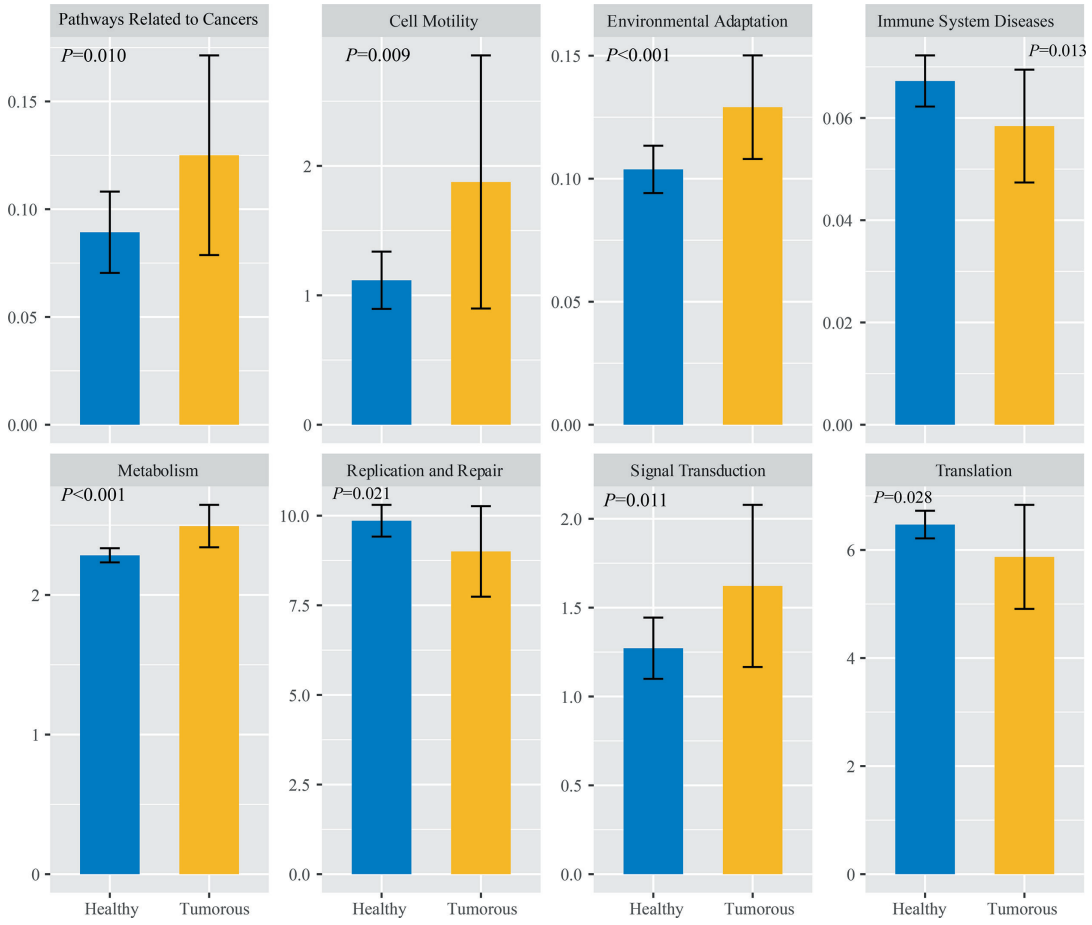


Fig. 6. The gene functions on level-2 that were significantly increased in different groups. The functions were predicted by PICRUSt with default settings, then the *t*-test and Storey FDR correction were conducted using STAMP. The functions connected with cancers (corrected *P*=0.010), cell motility (corrected *P*=0.009), environmental adaptation (corrected *P*<0.001), metabolism (corrected *P*<0.001) and signal transduction (corrected *P*<0.011) were significantly enhanced in the tumorous group compared with the healthy control group. However, the pathways related to immune system diseases (corrected *P*=0.013), replication and repair (corrected *P*=0.021), and translation (corrected *P*=0.028) were significantly enhanced in the healthy control group.

outcomes of eight diverse malignancy types and found that a more diverse intestinal microbiome was associated with improved systemic and antitumour immunity [47]. Preserving diversity is an important element of the survival of various microbiota [48]. Changes in the oral microenvironment in tumorous patients may contribute to the changes in diversity, which is more likely to induce a favourable antitumour response. However, some authors reported a significant reduction in microbial diversity in cancer patients compared with controls [39]. This discrepancy may be attributable to the different ethnic background or lifestyle-related factors in the study population. The relationship between microbial diversity and oral disease status needs to be identified.

In this study, *Proteobacteria*, *Bacteroidetes*, *Firmicutes*, *Fusobacteria* and *Actinobacteria* were the most abundant phyla (Fig. 2). These results are similar to those of other studies [40]. However, the most abundant bacteria in these phyla by percentage were not consistent with this study. *Proteobacteria* was the most abundant phylum detected in our study, while *Bacteroidetes* was the most abundant phylum reported by Zhao *et al.* [40]. *Firmicutes* was the most abundant phylum reported in another study [21]. *Neisseria* was the most abundant genus in our study, while *Prevotella* was the most abundant genus reported in the previous study [40] (Fig. 3). These differences may be attributable to the different genetic characteristics of the population or the different timing of sampling; therefore, more accurate standardized inclusion criteria and protocols are required to achieve consistent results.

In our study, *Neisseria*, *Prevotella*, *Streptococcus*, *Veillonella*, *Fusobacterium*, *Capnocytophaga* and *Haemophilus* were the most abundant genera; all these have been regarded as the common core oral microbiota in other studies [49]. *Neisseria* was the most abundant genus in our healthy control group, while *Prevotella* was the most abundant in the tumorous group. Yoshihisa *et al.* suggested that the predominance of the genus *Prevotella* in the salivary microbiota could be attributed to severe periodontal disease, while the predominance of the genus *Neisseria* indicated healthy periodontal conditions [38]. Similar results were reported

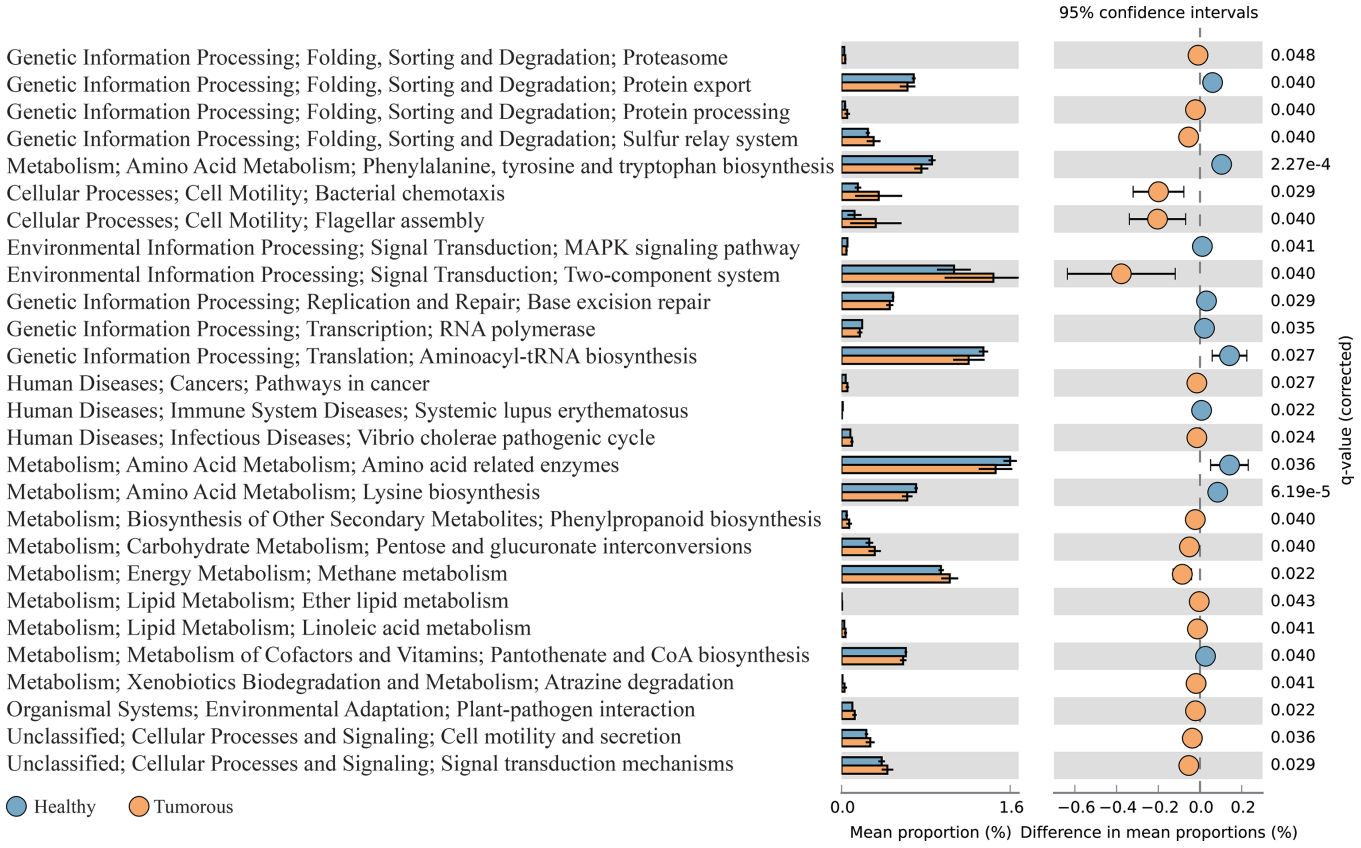


Fig. 7. The functions on level-3 that were significantly increased in different groups. The functions were predicted by PICRUSt with default settings, then the *t*-test and Storey FDR correction were conducted using STAMP. Twenty-seven different pathways were significantly different. Among these, 17 were more abundant in the tumorous group and 10 were were more abundant in the control healthy group (*t*-test, Storey FDR correction, corrected $P<0.05$). The pathways that were related to mitogen-activated protein kinase (MAPK) signalling, two-component system, bacterial chemotaxis, flagellar assembly, sulphur relay systems and cell motility were significantly increased in the tumorous group compared with the healthy control group. However, the samples from the control group had more genes related to lysine biosynthesis and aromatic amino acids (phenylalanine, tyrosine and tryptophan) biosynthesis.

by Rafael *et al.*, who compared the salivary microbiome from oropharyngeal cancer (OPSCC), OCSCC and normal healthy controls, and found a significant decrease in *Neisseria* in OPSCC and OCSCC [39]. Zhao *et al.* suggested that *Neisseria* may play an important role in maintaining the stability of the oral bacterial community [40]. In addition, our study found significantly greater abundance of *Streptococcus* in the healthy group. Previous studies have suggested that *Streptococcus* may be beneficial to the host, because these bacteria may produce molecules that are inhibitory to pathogenic species [50]. Tagaino *et al.* suggested that *Streptococcus mitis*, *Streptococcus salivarius* and *Streptococcus mutans* may have the ability to produce acetaldehyde from ethanol or glucose [51]. Acetaldehyde is a carcinogen and was shown to contribute to sister chromatid exchanges, gene mutations and DNA-strand breaks [51, 52]. The production of acetaldehyde is affected by the oral microenvironment, such as pH, oxygen concentration, metabolic substrates and even salivary secretions [51]. Therefore, the production of acetaldehyde in healthy individuals should be studied in future research.

Of the top 30 genera, *Sphingobium* was almost only exclusively discovered in healthy individuals. Recent studies have suggested its potential protective role in breast cancer, because of its ability to metabolize oestrogen [53, 54]. *Aggregatibacter* was more abundant in tumorous patients in our study. *Aggregatibacter actinomycetemcomitans* is one of the key causative pathogens in chronic periodontitis. In addition, it has been associated with aggressive periodontitis [55]. Long-term chronic periodontal disease increases the risk of oral cavity cancers, especially OSCC [56]. However, consistent chronic inflammation may not be the only reason for carcinogenesis, because accumulation of genome instabilities, such as point mutations, deletions, inversions, duplications, translocations and chromosome loss, is a key factor for carcinogenesis [57]. Teshima *et al.* discovered that *Aggregatibacter actinomycetemcomitans* infection can cause breakage in the DNA double-strand in host cells [57], which suggested that *Aggregatibacter actinomycetemcomitans* may increase the risk of genome instabilities in host cells and subsequently increase the risk of carcinogenesis. Leukotoxin (LtxA), which is produced by *Aggregatibacter actinomycetemcomitans,* can kill neutrophils,

lymphocytes and monocytes, allowing it to evade host immune surveillance [58]. Therefore, *Aggregatibacter actinomycetemcomitans* may have a deleterious effect on the human immune system. In addition, it has been shown to be associated with systemic complications, such as endocarditis, rheumatoid arthritis [59] and lung cancer [60, 61]. Some studies have found an association between *Aggregatibacter actinomycetemcomitans* and development of pancreatic cancer and oesophageal cancer [62, 63]. Therefore, further research is required to reveal its role in the development and progression of oral cancers. In addition, *Acinetobacter baumannii* is one of the most common clinical multidrug-resistant bacteria that is frequently isolated in wound infection following surgical intervention for OSCC [64]. However, our study was conducted before the surgical procedure; therefore, higher numbers of *Acinetobacter baumannii* may have existed before surgery, which might have contributed to its commonly reported isolation following oral cancer surgery. Inhibiting *Acinetobacter* growth before a surgical procedure could be an effective method to reduce multidrug-resistant bacterial infection. The abundance of the genus *Propionibacterium* was significantly greater in the tumour group compared with that in the healthy control group. Perera *et al.* discovered that rates were higher in fibroepithelial polyps (FEPs) compared with OSCCs [65]. It would be of interest to explore the role of *Propionibacterium* in the aetiology of FEPs or benign tumours.

PICRUSt and the STAMP method were used to assess the functions of various bacterial communities. Genes related to signal transduction were upregulated in the tumorous group compared with the healthy controls (Fig. 6). Further analysis suggested that the genes connected to two-component sensing systems (TCSs) were more abundant in the tumorous group (1.46–1.08%, P=0.04, t-test, FDR-corrected). These regulatory systems are ubiquitous signalling units found in prokaryotes, which tailor the response based on the local conditions. TCSs were shown to mediate bacterial acclimatization to various environmental changes by coupling environmental signals to gene expression [66]. Upregulation of these systems may promote bacterial survival. The correlation of MAPK pathway changes with the oral microbiota may inform novel combination therapies for oral tumours. Of interest, bacterial chemotaxis and flagellar assembly, which are involved in cell motility, were significantly higher in the tumorous group (Fig. 7). Williams *et al.* showed that *H. pylori* mutants with defective chemotaxis induced less inflammation than the wild-type *H. pylori* [67]. Wang *et al.* suggested an important role of bacterial chemotaxis in periodontitis-related inflammation [68]. Similar results were indicated in other reports [18]. Flagellin was shown to stimulate significant migration of innate immune cells, such as macrophages, dendritic cells and neutrophils to colonized tumours [69]. The functions of both genes demonstrate their antitumour ability, and the upregulation of these suggested antitumour immunological effects of some bacteria in the oral microbiota. In addition, genes correlated with replication and repair were significantly reduced in tumorous patients. Oral bacterial virulence factors produced by bacteria such as *Aggregatibacter actinomycetemcomitans* (which was more abundant in the tumorous group, Fig. 5b) may induce single-strand DNA breakages, produce replication stress response, and ultimately lead to breakage to double-strands and arrest cell cycle replication [70]. Inefficient repair of damaged DNA could allow survival of cells with defective DNA, resulting in accumulation of multiple genome instabilities and promoting tumorigenesis and progression [71]. In addition, the pathways related to immune system disease, replication or repair, and translation were significantly overrepresented in controls. Lysine biosynthesis was upregulated in healthy controls (Fig. 7). Because the bacterial biosynthesis of lysine is important in the construction of bacterial peptidoglycan cell walls [72], the underrepresented lysine biosynthesis genes in tumours indicated that their microenvironment had compromised the microbiota. Biosynthesis of three aromatic amino acids (phenylalanine, tyrosine and tryptophan) was upregulated in healthy controls (Fig. 7). These aromatic amino acids are essential for protein synthesis [73], and dysmetabolism of aromatic amino acids leads to cancers [74].

Some limitations of our study should be considered. First, the small sample size may have reduced the number of significantly different genera between the tumorous and healthy groups. However, 21 genera were identified, which included some genera that have been reported by other researchers. Second, consumption of alcohol and tobacco smoking are important aetiological factors for oral cancers, and therefore these factors are likely to affect the oral microbiota. Because of the small sample size, we did not discuss their impact. Therefore, large samples and a detailed comprehensive questionnaire survey on living habits might be required in future studies. Third, we did not assess the variations in the significantly different genera during cancer progression from early to advanced stages. Further validation of the statistically different genera in various histopathological stages of dysplastic oral cancer could provide valuable insights.

In conclusion, we observed a correlation between different variations in the oral microbiota and its functions with oral tumours. The tumorous group exhibited significantly greater diversity, which may indicate poor oral health. The tumorous group also showed significant overrepresentation of the abundance of some multi-drug-resistant and periodontitis-related pathogens. The significantly altered microbiota could be explored further as novel biomarkers and may help inform combination therapies for oral tumours. The results of this study expended our knowledge of oral bacterial communities in patients with oral tumours.

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Author contributions
J.Y. and X.W. conceived and designed research; M.Z., J.Z., X.T. and A.W. collected data and conducted research; P.H. and S.L. analysed and interpreted data; P.H. and J.Y. wrote the initial paper; X.W. and P.H. revised the paper; X.W. had primary responsibility for final content. All authors read and approved the final manuscript.

Conflicts of interest
The authors declare that they have no conflicts of interest.

Ethical statement
This work was approved by the Research Ethics Committee of the Hospital of Stomatology, Sun Yat-sen University at District of Yuexiu, Guangzhou City, China.

References

1. Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, *et al*. Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 2018;68:394–424.
2. Kademani D. Oral cancer. *Mayo Clin Proc* 2007;82:878–887.
3. Siegel RL, Miller KD, Jemal A. Cancer statistics, 2020. *CA Cancer J Clin* 2020;70:7–30.
4. Huber MA, Tantiwongkosi B. Oral and oropharyngeal cancer. *Med Clin North Am* 2014;98:1299–1321.
5. Fu J-Y, Wu C-X, Zhang C-P, Gao J, Luo J-F, *et al*. Oral cancer incidence in Shanghai ---- a temporal trend analysis from 2003 to 2012. *BMC Cancer* 2018;18:686.
6. Lei L, Zheng R, Peng K, Si L, Peng J, *et al*. Incidence and mortality of oral and oropharyngeal cancer in China, 2015. *Chin J Cancer Res* 2020;32:1–9.
7. Lin X, Wu X, Gomaa A, Chen J, Wu L, *et al*. Analysis of risk factors for multiple primary oral squamous cell carcinoma: a cohort study. *Clin Oral Investig* 2020;24:3147–3155.
8. Guha N, Warnakulasuriya S, Vlaanderen J, Straif K. Betel quid chewing and the risk of oral and oropharyngeal cancers: a meta-analysis with implications for cancer control. *Int J Cancer* 2014;135:1433–1443.
9. Hung L-C, Kung P-T, Lung C-H, Tsai M-H, Liu S-A, *et al*. Assessment of the risk of oral cancer Incidence in a high-risk population and establishment of a predictive model for oral cancer incidence using a population-based cohort in Taiwan. *Int J Environ Res Public Health* 2020;17:E665.
10. Bai X-X, Zhang J, Wei L. Analysis of primary oral and oropharyngeal squamous cell carcinoma in inhabitants of Beijing, China-a 10-year continuous single-center study. *BMC Oral Health* 2020;20:208.
11. Kim SS, Ruiz VE, Carroll JD, Moss SF. *Helicobacter pylori* in the pathogenesis of gastric cancer and gastric lymphoma. *Cancer Lett* 2011;305:228–238.
12. Rochford R, Korir A, Newton R. Viral-associated malignancies in Africa: are viruses "infectious traces" or "dominant drivers"? *Curr Opin Virol* 2016;20:28–33.
13. Flavahan WA, Gaskell E, Bernstein BE. Epigenetic plasticity and the hallmarks of cancer. *Science* 2017;357:eaal2380.
14. Doorakkers E, Lagergren J, Engstrand L, Brusselaers N. *Helicobacter pylori* eradication treatment and the risk of gastric adenocarcinoma in a Western population. *Gut* 2018;67:2092–2096.
15. Baker JL, Bor B, Agnello M, Shi W, He X. Ecology of the oral microbiome: beyond bacteria. *Trends Microbiol* 2017;25:362–374.
16. Yang C-Y, Yeh Y-M, Yu H-Y, Chin C-Y, Hsu C-W, *et al*. Oral microbiota community dynamics associated with oral squamous cell carcinoma staging. *Front Microbiol* 2018;9:862.
17. Bolz J, Dosá E, Schubert J, Eckert AW. Bacterial colonization of microbial biofilms in oral squamous cell carcinoma. *Clin Oral Investig* 2014;18:409–414.
18. Al-Hebshi NN, Nasher AT, Maryoud MY, Homeida HE, Chen T, *et al*. Inflammatory bacteriome featuring *Fusobacterium nucleatum* and *Pseudomonas aeruginosa* identified in association with oral squamous cell carcinoma. *Sci Rep* 2017;7:1834.
19. Chen MY, Chen JW, Wu LW, Huang KC, Chen JY, *et al*. Carcinogenesis of male oral submucous fibrosis alters salivary microbiomes. *J Dent Res* 2021;100:397–405.
20. Lee W-H, Chen H-M, Yang S-F, Liang C, Peng C-Y, *et al*. Bacterial alterations in salivary microbiota and their association in oral cancer. *Sci Rep* 2017;7:16540.
21. Schmidt BL, Kuczynski J, Bhattacharya A, Huey B, Corby PM, *et al*. Changes in abundance of oral microbiota associated with oral cancer. *PLoS One* 2014;9:e98741.
22. Mai X, Genco RJ, LaMonte MJ, Hovey KM, Freudenheim JL, *et al*. Periodontal pathogens and risk of incident cancer in postmenopausal females: the buffalo osteoperio study. *J Periodontol* 2016;87:257–267.
23. Banerjee S, Tian T, Wei Z, Peck KN, Shih N, *et al*. Microbial signatures associated with oropharyngeal and oral squamous cell carcinomas. *Sci Rep* 2017;7:4036.
24. Komiya Y, Shimomura Y, Higurashi T, Sugi Y, Arimoto J, *et al*. Patients with colorectal cancer have identical strains of *Fusobacterium nucleatum* in their colorectal cancer and oral cavity. *Gut* 2019;68:1335–1337.
25. Gao S, Li S, Ma Z, Liang S, Shan T, *et al*. Presence of *Porphyromonas gingivalis* in esophagus and its association with the clinicopathological characteristics and survival in patients with esophageal cancer. *Infect Agent Cancer* 2016;11:3.
26. Gao S-G, Yang J-Q, Ma Z-K, Yuan X, Zhao C, *et al*. Preoperative serum immunoglobulin G and A antibodies to *Porphyromonas gingivalis* are potential serum biomarkers for the diagnosis and prognosis of esophageal squamous cell carcinoma. *BMC Cancer* 2018;18:17.
27. Castellarin M, Warren RL, Freeman JD, Dreolini L, Krzywinski M, *et al*. *Fusobacterium nucleatum* infection is prevalent in human colorectal carcinoma. *Genome Res* 2012;22:299–306.
28. Vartoukian SR, Palmer RM, Wade WG. Strategies for culture of "unculturable" bacteria. *FEMS Microbiol Lett* 2010;309:1–7.
29. Vo A-TE, Jedlicka JA. Protocols for metagenomic DNA extraction and Illumina amplicon library preparation for faecal and swab samples. *Mol Ecol Resour* 2014;14:1183–1197.
30. Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, *et al*. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol* 2019;37:852–857.
31. Kaehler BD, Bokulich NA, McDonald D, Knight R, Caporaso JG, *et al*. Species abundance information improves sequence taxonomy classification accuracy. *Nat Commun* 2019;10:4643.
32. Bokulich NA, Kaehler BD, Rideout JR, Dillon M, Bolyen E, *et al*. Optimizing taxonomic classification of marker-gene amplicon sequences with QIIME 2's q2-feature-classifier plugin. *Microbiome* 2018;6:90.
33. Ye SH, Siddle KJ, Park DJ, Sabeti PC. Benchmarking metagenomics tools for taxonomic classification. *Cell* 2019;178:779–794.
34. Langille MGI, Zaneveld J, Caporaso JG, McDonald D, Knights D, *et al*. Predictive functional profiling of microbial communities using 16S rRNA marker gene sequences. *Nat Biotechnol* 2013;31:814–821.
35. Sun J, Tang Q, Yu S, Xie M, Xie Y, *et al*. Role of the oral microbiota in cancer evolution and progression. *Cancer Med* 2020;9:6306–6321.

36. Deo PN, Deshmukh R. Oral microbiome: unveiling the fundamentals. *J Oral Maxillofac Pathol* 2019;23:122–128.
37. Chattopadhyay I, Verma M, Panda M. Role of oral microbiome signatures in diagnosis and prognosis of oral cancer. *Technol Cancer Res Treat* 2019;18:1533033819867354.
38. Yamashita Y, Takeshita T. The oral microbiome and human health. *J Oral Sci* 2017;59:201–206.
39. Guerrero-Preston R, Godoy-Vitorino F, Jedlicka A, Rodríguez-Hilario A, González H, *et al*. 16S rRNA amplicon sequencing identifies microbiota associated with oral cancer, human papilloma virus infection and surgical treatment. *Oncotarget* 2016;7:51320–51334.
40. Zhao H, Chu M, Huang Z, Yang X, Ran S, *et al*. Variations in oral microbiota associated with oral cancer. *Sci Rep* 2017;7:11773.
41. Wang L, Yin G, Guo Y, Zhao Y, Zhao M, *et al*. Variations in oral microbiota composition are associated with a risk of throat cancer. *Front Cell Infect Microbiol* 2019;9:205.
42. Al-Hebshi NN, Nasher AT, Idris AM, Chen T. Robust species taxonomy assignment algorithm for 16S rRNA NGS reads: application to oral carcinoma samples. *J Oral Microbiol* 2015;7:28934.
43. Sun DL, Jiang X, Wu QL, Zhou NY. Intragenomic heterogeneity of 16S rRNA genes causes overestimation of prokaryotic diversity. *Appl Environ Microbiol* 2013;79:5962–5969.
44. Li Y, Tan X, Zhao X, Xu Z, Dai W, *et al*. Composition and function of oral microbiota between gingival squamous cell carcinoma and periodontitis. *Oral Oncol* 2020;107:104710.
45. Takeshita T, Kageyama S, Furuta M, Tsuboi H, Takeuchi K, *et al*. Bacterial diversity in saliva and oral health-related conditions: the Hisayama Study. *Sci Rep* 2016;6:22164.
46. Sims TT, El Alam MB, Karpinets TV, Dorta-Estremera S, Hegde VL, *et al*. Gut microbiome diversity is an independent predictor of survival in cervical cancer patients receiving chemoradiation. *Commun Biol* 2021;4:237.
47. Heshiki Y, Vazquez-Uribe R, Li J, Ni Y, Quainoo S, *et al*. Predictable modulation of cancer treatment outcomes by the gut microbiota. *Microbiome* 2020;8:28.
48. Bhushan B, Yadav AP, Singh SB, Ganju L. Diversity and functional analysis of salivary microflora of Indian Antarctic expeditionaries. *J Oral Microbiol* 2019;11:1581513.
49. Zaura E, Keijser BJF, Huse SM, Crielaard W. Defining the healthy "core microbiome" of oral microbial communities. *BMC Microbiol* 2009;9:259.
50. Abranches J, Zeng L, Kajfasz JK, Palmer SR, Chakraborty B, *et al*. Biology of oral *Streptococci*. *Microbiol Spectr* 2018;6.
51. Tagaino R, Washio J, Abiko Y, Tanda N, Sasaki K, *et al*. Metabolic property of acetaldehyde production from ethanol and glucose by oral *Streptococcus* and *Neisseria*. *Sci Rep* 2019;9:10446.
52. Singh NP, Khan A. Acetaldehyde: genotoxicity and cytotoxicity in human lymphocytes. *Mutat Res* 1995;337:9–17.
53. Parida S, Sharma D. The power of small changes: comprehensive analyses of microbial dysbiosis in breast cancer. *Biochim Biophys Acta Rev Cancer* 2019;1871:392–405.
54. Lawani-Luwaji EU, Alade T. Sphingomonadaceae: protective against breast cancer? *Bull Natl Res Cent* 2020;44:191.
55. Velusamy SK, Sampathkumar V, Ramasubbu N, Paster BJ, Fine DH. *Aggregatibacter actinomycetemcomitans* colonization and persistence in a primate model. *Proc Natl Acad Sci U S A* 2019;116:22307–22313.
56. Tezal M, Sullivan MA, Hyland A, Marshall JR, Stoler D, *et al*. Chronic periodontitis and the incidence of head and neck squamous cell carcinoma. *Cancer Epidemiol Biomarkers Prev* 2009;18:2406–2412.
57. Teshima R, Hanada K, Akada J, Kawano K, Yamaoka Y. *Aggregatibacter actinomycetemcomitans* infection causes DNA double-strand breaks in host cells. *Genes Cells* 2018;23:264–273.
58. Sampathkumar V, Velusamy SK, Godboley D, Fine DH. Increased leukotoxin production: characterization of 100 base pairs within the 530 base pair leukotoxin promoter region of *Aggregatibacter actinomycetemcomitans*. *Sci Rep* 2017;7:1887.
59. Mukherjee A, Jantsch V, Khan R, Hartung W, Fischer R, *et al*. Rheumatoid arthritis-associated autoimmunity due to *Aggregatibacter actinomycetemcomitans* and its resolution with antibiotic therapy. *Front Immunol* 2018;9:2352.
60. Turner E, Hastie T, Sundaresan PD. Pulmonary infection masquerades as malignancy in a patient with periodontitis. *BMJ Case Rep* 2021;14.
61. Matzumura-Kuan M, Jennings J. *Aggregatibacter actinomycetemcomitans* infection mimicking lung cancer: a case report. *Scand J Infect Dis* 2014;46:669–672.
62. Fan X, Alekseyenko AV, Wu J, Peters BA, Jacobs EJ, *et al*. Human oral microbiome and prospective risk for pancreatic cancer: a population-based nested case-control study. *Gut* 2018;67:120–127.
63. Kawasaki M, Ikeda Y, Ikeda E, Takahashi M, Tanaka D, *et al*. Oral infectious bacteria in dental plaque and saliva as risk factors in patients with esophageal cancer. *Cancer* 2021;127:512–519.
64. Belusic-Gobic M, Zubovic A, Predrijevac A, Harmicar D, Cerovic R, *et al*. Microbiology of wound infection after oral cancer surgery. *J Craniomaxillofac Surg* 2020;48:700–705.
65. Perera M, Al-Hebshi NN, Perera I, Ipe D, Ulett GC, *et al*. Inflammatory bacteriome and oral squamous cell carcinoma. *J Dent Res* 2018;97:725–732.
66. Ibrahim IM, Puthiyaveetil S, Allen JF. A Two-component regulatory system in transcriptional control of photosystem stoichiometry: redox-dependent and sodium ion-dependent phosphoryl transfer from cyanobacterial histidine kinase hik2 to response regulators Rre1 and RppA. *Front Plant Sci* 2016;7:137.
67. Williams SM, Chen YT, Andermann TM, Carter JE, McGee DJ, *et al*. *Helicobacter pylori* chemotaxis modulates inflammation and bacterium-gastric epithelium interactions in infected mice. *Infect Immun* 2007;75:3747–3757.
68. Wang J, Qi J, Zhao H, He S, Zhang Y, *et al*. Metagenomic sequencing reveals microbiota and its functional potential associated with periodontal disease. *Sci Rep* 2013;3:1843.
69. Duong MT-Q, Qin Y, You S-H, Min J-J. Bacteria-cancer interactions: bacteria-based cancer therapy. *Exp Mol Med* 2019;51:1–15.
70. Sun Z, Xiong C, Teh SW, Lim JCW, Kumar S, *et al*. Mechanisms of oral bacterial virulence factors in pancreatic cancer. *Front Cell Infect Microbiol* 2019;9:412.
71. Negrini S, Gorgoulis VG, Halazonetis TD. Genomic instability--an evolving hallmark of cancer. *Nat Rev Mol Cell Biol* 2010;11:220–228.
72. Hutton CA, Perugini MA, Gerrard JA. Inhibition of lysine biosynthesis: an evolving antibiotic strategy. *Mol Biosyst* 2007;3:458–465.
73. Ennis MA, Ong A-J, Lim K, Ball RO, Pencharz PB, *et al*. Dietary aromatic amino acid requirements during early and late gestation in healthy pregnant women. *J Nutr* 2020;150:3224–3230.
74. Sitkin SI, Tkachenko EI, Vakhitov TY. Metabolic dysbiosis of the gut microbiota and its biomarkers. *Eksp Klin Gastroenterol* 2016;12:6–29.